



DO-IT-YOURSELF

Simple Multi-Sensor FIRE ALARM

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The circuit presented here raises an alarm on detecting flames and high temperatures. It can be used in two ways: A single sensor board with an alarm board, or three sensor boards with an alarm board. The three separate sensor boards are to be installed at three different places, with the alarm unit installed at an appropriate location.

Circuit and working

Circuit diagram of the fire alarm is shown in Fig. 1. It is built around a transformer (X1), 6V voltage regulator 7806 (IC1), temperature sensor LM35 (IC2), flame sensor IR LED (IRLED1), op-amp LM358 (IC3), transistors BC547 and BC557 (T1 and T2), a 6V single-changeover relay and a few other components.

The circuit senses ambient temperature using LM35 and checks the

presence of flame with an IR sensor. Yellow- or red-coloured flames emit IR radiation, which is sensed by the IR sensor (IRLED1).

In case yellow/red flame, high temperature or both are detected, dual comparator LM358 triggers npn transistor T1. Transistor T1 conducts and its collector goes low to drive pnp transistor T2 into conduction. This, in turn, energises relay RL1 to activate the load (buzzer, lights and other equipment) connected at CON9.

One big advantage of this circuit is that in case the sensor board loses connection from the alarm board due to fire or explosion, transistor T2 gets activated due to logic low at its base and thus still energises the relay. So, place the alarm board at an appropriate remote location for monitoring.

Normally, when there is no fire, LED3 and LED4 on the sensor board are 'off.' LED4 glows when the temperature crosses the set temperature range, while LED3 glows when the IR sensor detects IR radiation from flame. LED5 glows during normal operation and goes off when it receives a signal from the sensor board.

The output of the sensor board at connector CON5 is marked as 'A'.

Three similar sensor boards at different locations can be used by connecting their respective outputs to CON6 through CON8 of the alarm board. This arrangement is useful for monitoring multiple rooms, corridors, hotel rooms and staircases where alarm

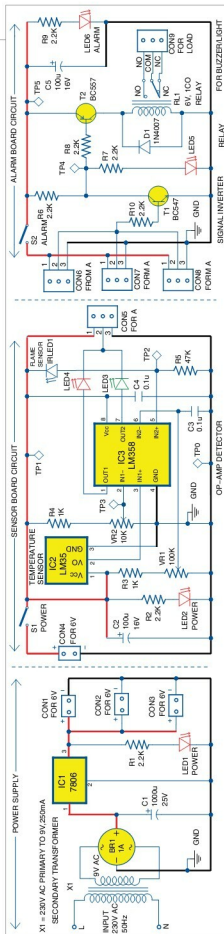


Fig. 1: Circuit diagram of fire alarm

PARTS LIST

Semiconductors:

- 7806, 6V regulator
- IC2 - LM35 temperature sensor
- IC3 - LM358 op-amp

IRLED1

- IR LED flame sensor
- T1 - BC547 npn transistor
- T2 - BC557 pnp transistor

LED1-LED6

- 5mm LED
- D1 - 1N4007 rectifier diode

BR1

- Bridge rectifier

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R6-R10 - 2.2-kilo-ohm
- R3, R4 - 1-kilo-ohm
- R5 - 47-kilo-ohm
- VR1 - 100-kilo-ohm potmeter
- VR2 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2, C5 - 100 μ F, 16V electrolytic
- C3, C4 - 0.1 μ F ceramic disk

Miscellaneous:

- CON1-CON4 - 2-pin connector
- CON5-CON9 - 3-pin connector
- S1, S2 - On/off switch
- RL1 - 6V, 1CO relay
- X1 - 230V AC primary to 9V, 250mA secondary transformer
- 230V AC/50Hz or 6V battery